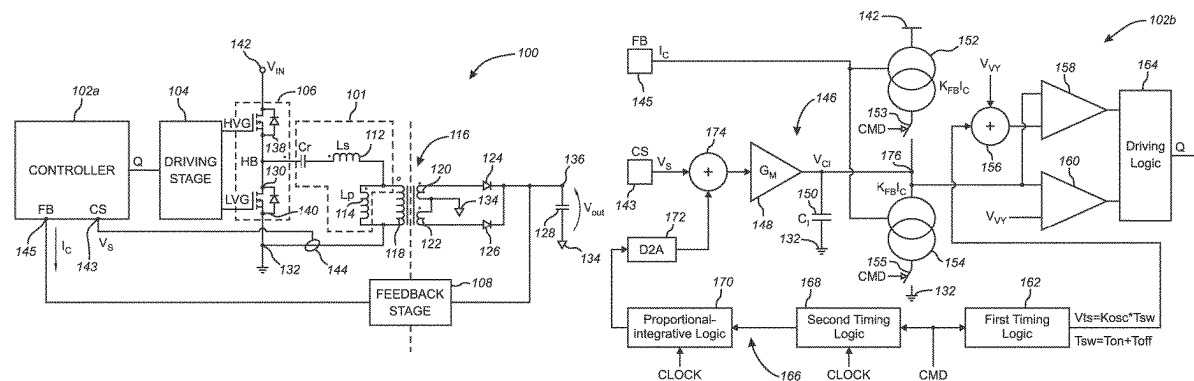


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**20 Claims, 3 Drawing Sheets**



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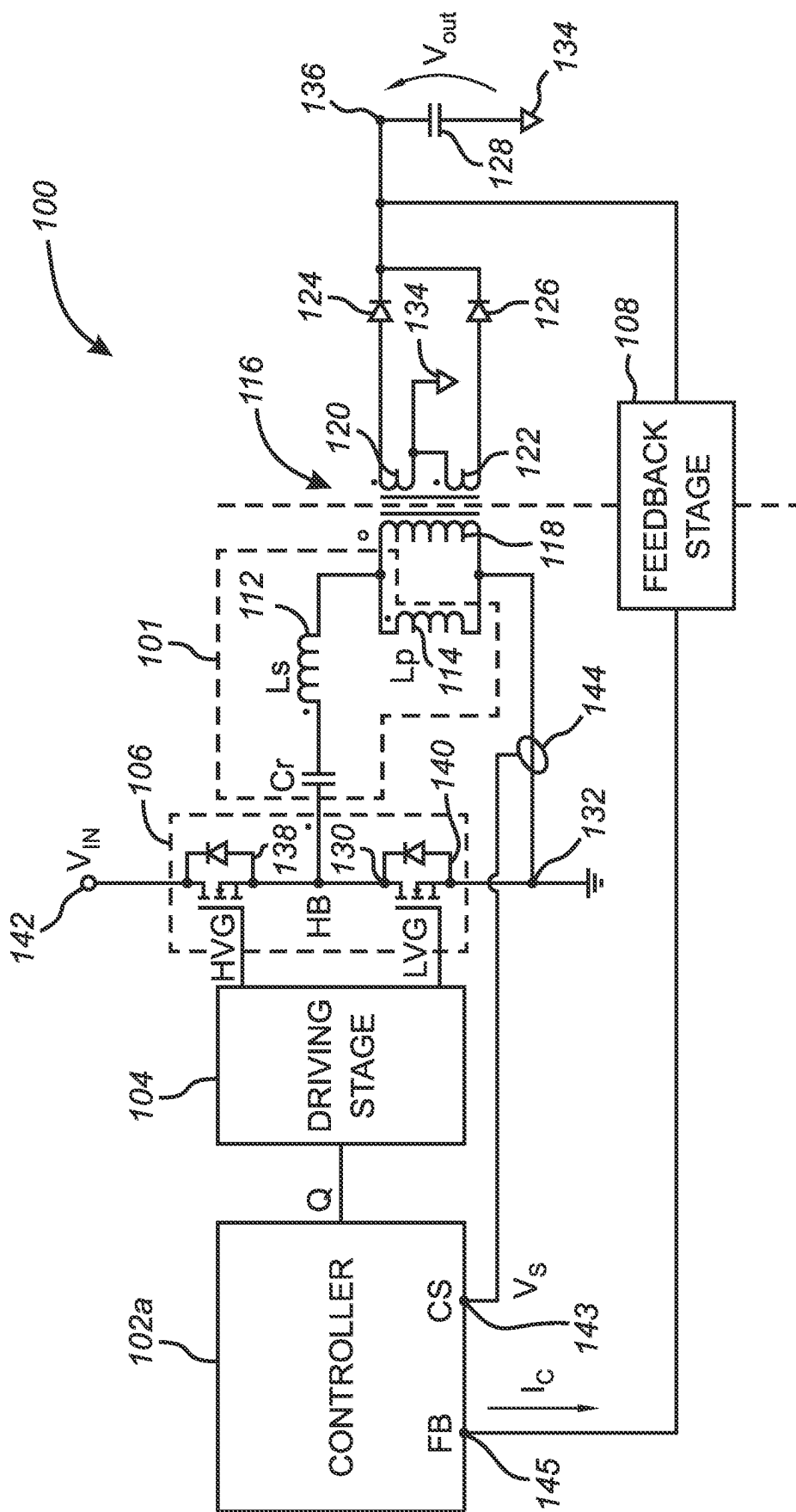


FIG. 1

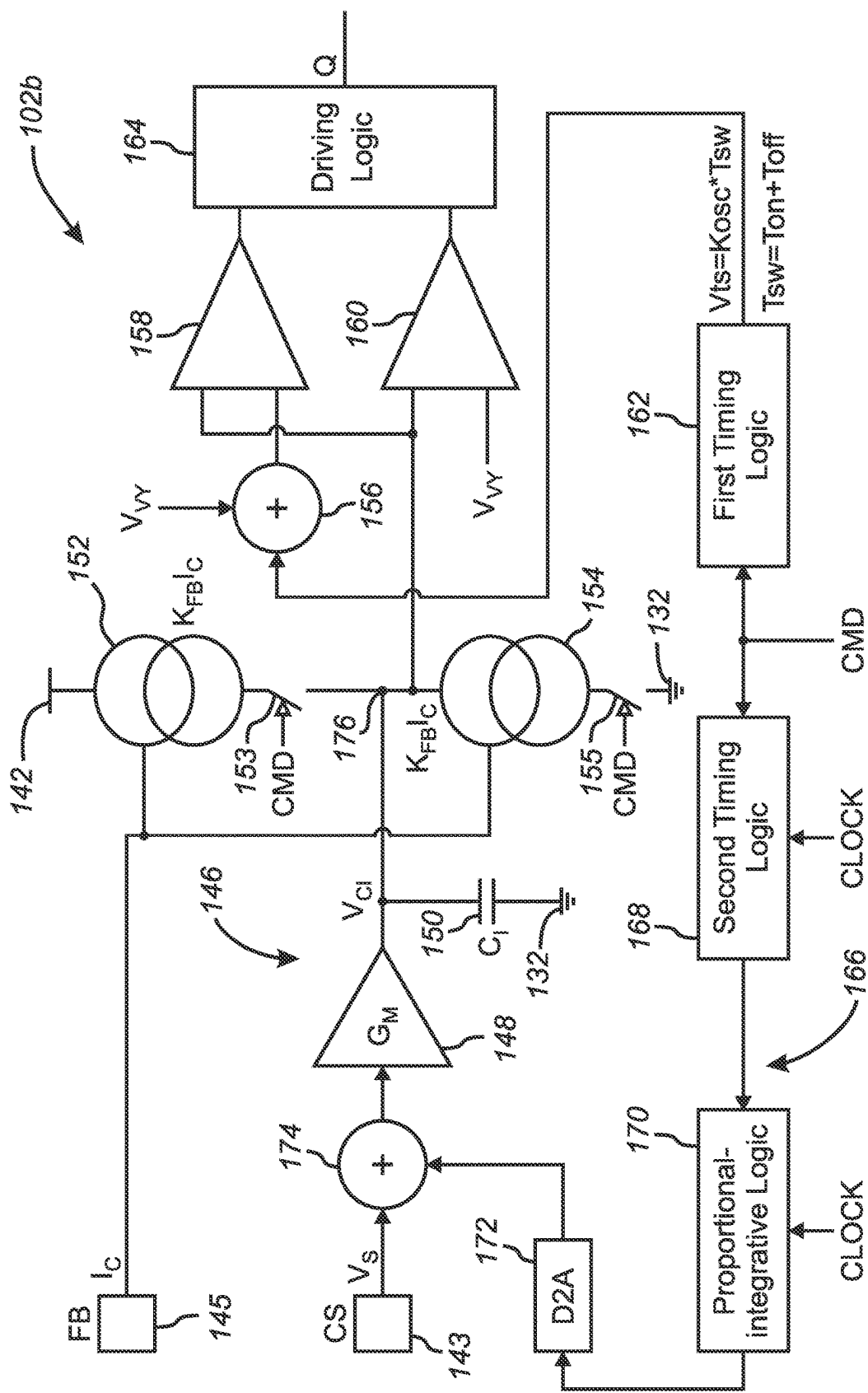
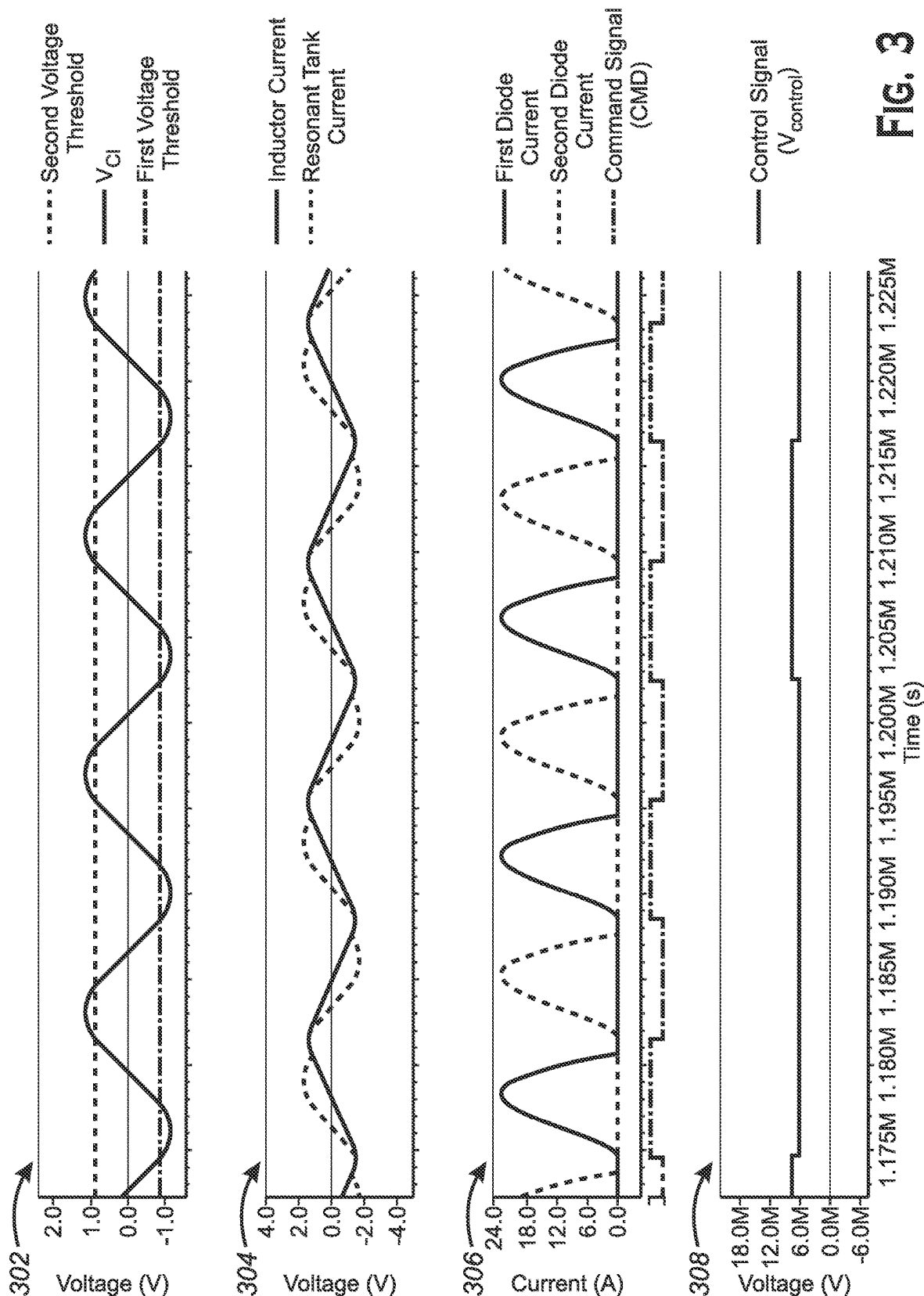


FIG. 2



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## CHARGE CONTROL FOR RESONANT CONVERTERS

### BACKGROUND

#### Technical Field

The present disclosure is directed to controlling a converter and, in particular, controlling the converter to mitigate current imbalance in the converter.

#### Description of the Related Art

A resonant converter is typically controlled by feeding back a signal representative of an output voltage of the converter and using the signal to control a switching of the converter. The primary and secondary sides of the converter are typically galvanically isolated using a transformer and an optocoupler. In particular, the signal representative of the output voltage of the converter is fed back through the optocoupler to maintain galvanic isolation between the primary and secondary sides.

### BRIEF SUMMARY

A converter may employ two diodes on the converter's secondary side, whereby the two diodes may be used to supply current to a load. It is desirable for the currents passing through the diodes to be symmetric. The currents may be symmetric when the currents largely correspond to each other, but are set apart from each other by a phase. The currents may become asymmetric due to artifacts in the controller. For example, the currents of the diodes may become asymmetric when there is an imbalance in the currents of the controller or when an offset is introduced in the controller.

The controller mitigates the asymmetry by determining a difference between an on-time of a signal representative of the switching of the converter and an off-time of the signal. The controller then compensates a feedback signal using the difference. The controller may compensate a feedback signal representative of a current in the resonant tank of the converter using the difference. The compensation may mitigate the diode current asymmetry.

### BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. 1 shows a circuit diagram of an LLC resonant converter coupled to control stages of the converter.

FIG. 2 show a controller of the converter in accordance with an embodiment.

FIG. 3 shows diagrams of signals of the converter.

### DETAILED DESCRIPTION

FIG. 1 shows a circuit diagram of an LLC resonant converter **100** coupled to control stages of the converter **100**. The control stages include a controller **102a**, a driving stage **104** and a feedback stage **108**. The LLC resonant converter **100** includes a switching stage **106** and a resonant tank **101** having two inductances (a resonant inductance **112** (denoted as 'Ls') and a shunt inductance **114** (denoted as 'Lp')) and one capacitance (a resonant capacitance **110** (denoted as 'Cr')). The converter **100** also includes a transformer **116** having a primary winding **118** in a primary side of the converter **100** (and the transformer **116**). The primary wind-

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ing **118** and the primary side are galvanically isolated from two secondary windings of a secondary side. The two secondary windings include a first secondary winding **120** and a second secondary winding **122**. The converter **100** also includes a first diode **124**, a second diode **126** and an output capacitance **128**.

The resonant capacitance **110** has a first side coupled to a half bridge node **130** (denoted as 'HB') of the switching stage **106**. The resonant capacitance **110** has a second side coupled to a first terminal of the resonant inductance **112**. The resonant inductance **112** has a second terminal coupled to both a first terminal of the shunt inductance **114** and a first terminal of the primary winding **118**. The shunt inductance **114** and the primary winding **118** each have second terminals that are coupled to each other and together coupled to a primary side reference voltage node **132**. The primary side reference voltage node **132** may be a ground node of the primary side.

The transformer's **116** secondary side is center-tapped. The first secondary winding **120** and the second secondary winding **122** each have first terminals that are coupled to each other and to a secondary side reference voltage node **134**. The first secondary winding **120** has a second terminal coupled to an anode of the first diode **124**. The second secondary winding **122** has a second terminal coupled to an anode of the second diode **126**. The first and second diodes **124**, **126** each have cathodes that are coupled to an output voltage node **136**. The output capacitance **128** is coupled between the output voltage node **136** and the secondary side reference voltage node **134**, which may be a ground node.

The feedback stage **108**, which may be an isolated feedback stage, such as an optocoupler, has an input coupled to the output voltage node **136** and an output coupled to the controller **102a**. The feedback stage **108** outputs, to the primary side, a signal (current  $I_c$ ) representative of the output voltage while maintaining the galvanic isolation between the primary and secondary sides.

The switching stage **106** includes a first transistor **138** and a second transistor **140**. Although the first and second transistors **138**, **140** are shown as n-channel metal-oxide semiconductor field-effect transistors (MOSFETs), it is noted that the first and second transistors **138**, **140** may be any other type of transistor or switch. The first transistor **138** has a source coupled to an input voltage node **142**, a drain coupled to the half bridge node **130** and a gate for receiving a high side gate drive signal (denoted as 'HVG') from the driving stage **104**. The input voltage node **142** supplies an input voltage (VIN) to the converter **100** via the first transistor **138**. The second transistor **140** has a source coupled to the half bridge node **130**, a drain coupled to the primary side reference voltage node **132** and a gate for receiving a low side gate drive signal (denoted as 'LVG') from the driving stage **104**.

The driving stage **104** has an input coupled to an output of the controller **102a**. The driving stage receives a switching signal (Q) from the controller **102a**. The switching signal has on and off states, which are also referred to as activated and deactivated states, respectively, and asserted and deasserted states, respectively. The driving stage **104** determines the states of the high side gate drive signal and the low side gate drive signal based on the switching signal received from the controller **102a**. The driving stage **104** then generates the high side and low side gate drive signals and outputs the high side and low side gate drive signals to the converter **100**.

The controller **102a** generates the switching signal (Q) based on various feedback signals. The control techniques

described herein use one or more of the plurality of feedback signals for controlling the converter 100. The controller 102a receives at a second input (FB) 145 the current ( $I_C$ ) representative of the output voltage ( $V_{OUT}$ ) of the converter 100. The controller 102a is also coupled to a primary side current sensing stage 144. The primary side current sensing stage 144 may include a sense resistance coupled to the primary side reference voltage node 132. The controller receives a voltage (referred to as  $V_S$ ) representative of the primary side current or the current of the resonant tank 101 at a first input (CS) 143 of the controller 102a. The voltage ( $V_S$ ) representative of the primary side current or the current of the resonant tank 101 may be available externally to the controller 102a. The controller 102a controls operation of the converter 100 based on the current representative of the output voltage and the voltage representative of the primary side current.

It is noted that although the current ( $I_C$ ) is described herein as being representative of the output voltage ( $V_{OUT}$ ) of the converter 100, in various embodiments, the current ( $I_C$ ) may be representative of the output current ( $I_{OUT}$ ) of the converter 100. In some applications, such light emitting diodes (LEDs), the converter 100 delivers a regulated output current to a load rather than a regulated output voltage ( $V_{OUT}$ ). In such embodiments, the control loop operates to maintain a constant output current ( $I_{OUT}$ ), and the control current ( $I_C$ ) is, therefore, representative of the output current ( $I_{OUT}$ ). The current ( $I_C$ ) may be representative of either the output voltage or the output current (i.e., whichever output parameter is to be controlled).

It is noted that sensing may also be carried out in other ways, such as through a capacitive or resistive divider, or a current transformer or Hall sensor, among others, to provide a voltage signal accurately representative of the instantaneous current flowing in the resonant tank circuit.

The controller 102a generates the switching signal (Q) based on one or more of the feedback signals. The controller 102a outputs switching signal to the driving stage 104, which in turn generates the high side and low side gate drive signals based on the switching signal. The high side and low side gate drive signals are in phase opposition, whereby if the high side gate drive signal (HVG) is asserted, the low side gate drive signal (LVG) is deasserted and vice-versa. The controller 102a may assert the high side gate drive signal (HVG) (and deassert the low side gate drive signal (LVG)) when the switching signal (Q) is asserted. Conversely, the controller 102a may deassert the high side gate drive signal (HVG) (and assert the low side gate drive signal (LVG)) when the switching signal (Q) is deasserted.

When the high side gate drive signal is asserted, the input voltage (VIN) is supplied to the first side of the resonant capacitance 110 to drive the resonant tank 101. When the low side gate drive signal is asserted, the primary side reference voltage, which may be ground voltage, is supplied to the first side of the resonant capacitance 110 to drive the resonant tank 101. When the resonant tank 101 is driven, energy in the primary side is transferred by the transformer 116 to the secondary side.

In the secondary side, the first and second diodes 124, 126 perform full-wave rectification that produces the output voltage ( $V_{OUT}$ ) at the output voltage node 136. The output capacitance 128 buffers and stabilizes the output voltage ( $V_{OUT}$ ). A load (not shown) may be coupled between the output voltage node 136 and the secondary side reference voltage node 134. The load is thereby driven by the output voltage ( $V_{OUT}$ ).

It is noted that although the transformer 116 is shown to include two secondary winding 120, 122, in various embodiments other configurations of the transformer 116 (and the converter 100) may be used. For example, a single secondary winding may be used together with a full bridge diode rectification in the secondary side of the converter 100.

The controller 102a may employ one or more closed-loop control techniques. For example, the controller 102a may use direct frequency control (DCF), which controls the switching frequency as to accommodate load demands. Further, the controller 102a may use a charge control technique. The charge control technique may use dual-loop control. An inner loop controls switch commutation based on an intermediate variable and on sensed converter variables (such as, resonant tank voltage or current). The intermediate variable may be generated by an outer loop (or a voltage loop).

One example of charge control is Average Input Current Control (AICC). In AICC, various techniques may be used to drive the first and second transistors 138, 140 using the high side gate drive signal (HVG) and the low side gate drive signal (LVG), respectively. The controller 102a may perform symmetric control. The controller 102a may use an integrator and two comparators to drive the first and second transistors 138, 140 in a symmetrical and complementary manner.

Additionally or alternatively, the controller 102a may use a comparator to trigger the first transistor 138 to turn off using the high side gate drive signal (HVG). The controller 102a may measure the on-time duration of the high side gate drive signal (HVG) and set the on-time duration of the low side gate drive signal (LVG) to the on-time duration of the high side gate drive signal (HVG). For example, the controller 102a may use a timer circuit to measure or set the time.

Accordingly, the controller 102a may drive the first and second transistors 138, 140 to have equal and complementary on-time durations. Thus, the first and second transistors 138, 140 may commute in complementary manner. The controller 102a may insert or add a dead time between the on-time and off-time durations of the first and second transistors 138, 140 to avoid or mitigate cross-conduction. In addition, the on-time durations and the off-time durations of the first and second transistors 138, 140 may be the same.

FIG. 2 show a controller 102b of the converter 100 in accordance with an embodiment. The controller 102b includes an integrator 146. The integrator 146 includes a transconductance amplifier ( $G_M$ ) 148 and a capacitance ( $C_I$ ) 150. The controller 102b includes first and second current sources 152, 154, first and second switches 153, 155, a first adder 156, first and second comparators 158, 160, first timing logic 162 and driving logic 164. The controller 102b includes a current balancing stage 166. The current balancing stage 166 includes second timing logic 168, proportional-integrative logic 170, a digital-to-analog converter 172 and a second adder 174.

The second adder 174 has a first input coupled to the first input 143 of the controller 102b, a second input coupled to an output of the digital-to-analog converter 172 and an output. The transconductance amplifier 148 has an input coupled to the output of the second adder 174 and an output. The capacitance 150 has a first side coupled to the output of the transconductance amplifier 148 (at control node 176) and a second side coupled to the primary side reference voltage node 132.

The first current source 152 has an anode coupled to the input voltage node 142, a control terminal coupled to the

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second input **145** of the controller **102b** and a cathode. The first switch **153** has a first terminal coupled to the cathode of the first current sources **152**, a second terminal coupled to the control node **176** and a control terminal configured to receive a command signal (CMD).

The command signal (CMD) may be based on the switching signal (Q), the high side gate drive signal (HVG) or the low side gate drive signal (LVG). For example, the on-time of the command signal (CMD) may be the on-time of the high side gate drive signal (HVG) as decreased by a dead time. The off-time of the command signal (CMD) may be the on-time of the low side gate drive signal (LVG) as increased by the dead time. The on-time of the command signal (CMD) may be the on-time of the switching signal (Q) as decreased by a dead time, and the off-time of the command signal (CMD) may be the off-time of the switching signal (Q) as increased by the dead time.

The second switch **155** has a first terminal coupled to the primary side reference voltage node **132**, a control terminal configured to receive the command signal (CMD) and a second terminal. The second current source **152** has a cathode coupled to the second terminal of the second switch **155**, an anode coupled to the control node **176** and a control terminal configured to receive the command signal (CMD).

The first timing logic **162** has an input configured to receive the command signal (CMD) and an output configured to provide a voltage (Vts) representative of a switching period ( $T_{SW}$ ) of the command signal (CMD). The first adder **156** has a first input coupled to the output of the first timing logic **162**, a second input configured to receive a first threshold voltage ( $V_{VT}$ ) and an output configured to provide a second threshold voltage that is a sum of the voltage (Vts) representative of the switching period ( $T_{SW}$ ) and the first threshold voltage ( $V_{VT}$ ).

The first comparator **158** has a first input coupled to the control node **176**, a second input coupled to the output of the first adder **156** (and configured to receive the second threshold voltage) and an output. The second comparator **160** has a first input coupled to the control node **176**, a second input configured to receive the first threshold voltage ( $V_{VT}$ ) and an output. The driving logic **164** has first and second inputs coupled to the output of the first comparator **158** and the output of the second comparator **160**, respectively. The driving logic **164** has an output configured to output the switching signal (Q) to the driving stage **104**.

The second timing logic **168** has a first input configured to receive the command signal (CMD), a second input configured to receive a clock signal and an output configured to output a signal representative of a difference between an on-time (Ton) (on state time, activation time or assertion time) and an off-time (Toff) (off state time, deactivation time or deassertion time) of the command signal (CMD). The proportional-integrative logic **170** has an input configured to receive the signal representative of the difference between the on-time and the off-time and an output configured to output a signal representative of an integral of the difference. The digital-to-analog converter **172** has an input coupled to the output of the proportional-integrative logic **170** and an output coupled to the second input of the second adder **174**. The digital-to-analog converter **172** outputs a control signal ( $V_{control}$ ) to the second adder **174**.

During operation, the second adder **174** sums the voltage ( $V_s$ ) representative of the current of the resonant tank **101** and the control signal ( $V_{control}$ ). The integrator **146** integrates the sum of the current of the resonant tank **101** and the control signal ( $V_{control}$ ) and charges the control node **176** to a voltage ( $V_{CT}$ ) according to the integral. The current ( $I_C$ )

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representative of the output voltage ( $V_{OUT}$ ) is also fed to the control node **176** by operation of the first and second current sources **152**, **154** and the first and second switches **153**, **155**.

When the command signal (CMD) is asserted, the first switch **153** is closed and the second switch **155** is open. The first current source **152** is controlled to supply a current to the control node **176** that is a function of (e.g., proportional to) the current ( $I_C$ ) representative of the output voltage ( $V_{OUT}$ ). When the command signal (CMD) is deasserted, the first switch **153** is open and the second switch **155** is closed. The first current source **152** no longer supplies current to the control node **176**. Instead, the second current source **154** sinks current from the control node **176**. The second current source **154** is controlled by the current ( $I_C$ ) representative of the output voltage ( $V_{OUT}$ ), and the second current source **154** sinks current that is a function of (e.g., proportional to) the current ( $I_C$ ). The first and second current sources **152**, **154** provide either positive or negative gain depending on which of the first and second switches **153**, **155** is closed.

The first timing logic **162** determines the switching period ( $T_{SW}$ ), e.g., as a sum of the on-time (Ton) and the off-time (Toff) of the command signal (CMD). The first timing logic **162** outputs the voltage (Vts) representative of the switching period ( $T_{SW}$ ). The first timing logic **162** may output the voltage (Vts) as a product of the switching period ( $T_{SW}$ ) and a factor ( $K_{OSC}$ ), e.g.,  $Vts = K_{OSC} * T_{SW}$ .

The first adder **156** sets the second voltage threshold to be greater than the first threshold voltage by the voltage (Vts). The first comparator **158** compares the voltage ( $V_{CT}$ ) of the control node **176** with the second threshold voltage. The first comparator **158** asserts its output in response to the voltage ( $V_{CT}$ ) of the control node **176** exceeding the second threshold voltage and deasserts its output if the voltage ( $V_{CT}$ ) of the control node **176** is less than or equal to the second threshold voltage.

The second comparator **160** compares the voltage ( $V_{CT}$ ) of the control node **176** with the first threshold voltage. The second comparator **160** asserts its output in response to the voltage ( $V_{CT}$ ) of the control node **176** being less than the first threshold voltage and deasserts its output if the voltage ( $V_{CT}$ ) of the control node **176** is greater than or equal to the first threshold voltage.

The driving logic **164** sets the switching signal (Q) based on the outputs of the first and second comparators **158**, **160**. For example, the driving logic **164** may deassert the switching signal (Q) (to cause the high side gate drive signal (HVG) to become deasserted and transition the first transistor **138** to the non-conductive state) in response to the output of the first comparator **158** being asserted. The driving logic **164** may assert the switching signal (Q) (to cause the high side gate drive signal (HVG) to become asserted and transition the first transistor **138** to the conductive state) in response to the output of the second comparator **160** asserted.

In the current balancing stage **166**, the second timing logic **168** receives the command signal (CMD) and determines a difference between the on-time (Ton) and the off-time (Toff) of the command signal (CMD). The second timing logic **168** may receive a clock signal and determine the difference as a number of clock cycles of the clock signal. For example, the second timing logic **168** may increment a counter every clock cycle of the clock signal during the on-time (Ton) and decrement the counter every clock cycle of the clock signal during the off-time (Toff) or vice-versa. The second timing logic **168** may include a counter (e.g., a clocked up-down counter) and an accumulator. The second timing logic **168** may increment the counter during the on-time and decre-



ment the counter during the off-time. A final count of the counter may represent the difference. The second timing logic **168** may initialize the counter to zero at the end of the switching period. A positive count represents a longer on-time duration than an off-time duration and vice-versa. A count of zero represents equal durations of the on-time and off-time.

For a switching period indexed by an index k, the difference between the on-time ( $T_{on}$ ) and the off-time ( $T_{off}$ ) determined may be represented as:

$$\frac{T_{onk} - T_{offk}}{T_{clock}}, \quad \text{Equation (1)}$$

where  $T_{clock}$  is the clock duration.

The proportional-integrative logic **170** may be a proportional-integral (PI) controller. The proportional-integrative logic **170** receives the difference from the second timing logic **168**. The proportional-integrative logic **170** may generate a weighted combination of the difference for a current switching period and the differences of one or more previous switching periods. For example, when a switching period indexed by an index i precedes the switching period indexed by the index k, the output of the proportional-integrative logic **170** may be represented as:

$$\frac{V_{eq}(T_{on_i} - T_{off_i}) - \lambda(T_{on_k} - T_{off_k})}{T_{clock}} \quad \text{Equation (2)}$$

where  $\lambda$  is a factor or a multiplier and  $V_{eq}$  is a voltage value.

The output of the proportional-integrative logic **170** may be any linear combination of the cumulative difference between on-time and off-time of the command signal (CMD). The output of the proportional-integrative logic **170** may be a signed array of bits representing the proportional-integral contribution. In some embodiments, one or more least significant bits of the array may be discarded to achieve coarser (rather than finer) control.

The digital-to-analog converter **172** receives the output of the proportional-integrative logic **170**. The digital-to-analog converter **172** converts the output of the proportional-integrative logic **170**, which it represented digitally, to the control signal ( $V_{control}$ ) having an analog representation. The second adder **174** sums the voltage ( $V_S$ ) representative of the current of the resonant tank **101** and the control signal ( $V_{control}$ ).

Imbalance and a lack of symmetry between the on-time and the off-time of the high side gate drive signal (HVG) or the low side gate drive signal (LVG) results in uneven current distribution through the first and second diodes **124**, **126**. When the currents passing through the diodes **124**, in **126** are uneven or symmetric, the reliability of the converter decreases and power dissipation increases. The on-time/off-time imbalance may also be caused by the fact that the transconductance amplifier ( $G_M$ ) **148** has an offset at its input. Further, the imbalance may be caused by mismatches between the currents sourced by the first and second current sources **152**, **154**.

The current balancing stage **166** aids in causing the high side gate drive signal (HVG) or the low side gate drive signal (LVG) to be symmetric. The current balancing stage **166** aids in causing the on-times of the high side gate drive signal (HVG) and the low side gate drive signal (LVG) to be equal and the off-times of the high side gate drive signal

(HVG) and the low side gate drive signal (LVG) to be equal. The current balancing stage **166** mitigates imbalance or unequal currents in the first and second diodes **124**, **126**. The control technique mitigates the effect of converter or controller element mismatches. The current balancing stage **166** determines the difference between the on-time and off-time of the command signal (CMD). The current balancing stage **166** supplies the control signal ( $V_{control}$ ) to the input of the transconductance amplifier ( $G_M$ ) **148** in order to compensate for and rectify the difference and aid in making the on-time and off-time the same. The current balancing stage **166** adjusts the voltage ( $V_S$ ) representative of the current of the resonant tank **101** to cause the controller **102b** to increase or decrease the on-time and off-time to reduce the difference between the on-time and the off-time.

When there is no difference between the on-time and off-time, the control signal ( $V_{control}$ ) has a value of zero volts and the current balancing stage **166** does not alter operation of the controller **102b**. However, when the on-time exceeds the off-time, the control signal ( $V_{control}$ ) is additively combined with the voltage ( $V_S$ ) to reduce the on-time. Conversely, when the off-time exceeds the on-time, the control signal ( $V_{control}$ ) is additively combined with the voltage ( $V_S$ ) to increase the on-time.

FIG. 3 shows diagrams of signals of the converter **100**. A first diagram **302** shows the first and second voltage thresholds and the voltage ( $V_{CI}$ ) of the control node **176**. A second diagram **304** shows the voltage ( $V_S$ ) representative of the current of the resonant tank **101** and an inductor current of the converter **100**. A third diagram **306** shows the currents through the first and second diodes **124**, **126** and the command signal (CMD). A fourth diagram **308** shows the control signal ( $V_{control}$ ) of the current balancing stage **166**.

When the transconductance amplifier ( $G_M$ ) **148** has an offset at its input, the currents through the first and second diodes **124**, **126** will be mismatched. As shown in FIG. 3, the control signal ( $V_{control}$ ) converged at a voltage value alternating between 6 and 8 millivolts (mV) to compensate for the offset. The resulting currents through the first and second diodes **124**, **126** are symmetrical during operation of the converter **100**.

The techniques described herein may control any type of resonant converter based on a bang-bang control using two comparators and an integrator. In addition, although digital techniques are described herein, an analog technique may be used. For example, a first capacitance may be charged during the on-time and discharged during the off-time to determine the difference between the two time intervals. A voltage of the first capacitance may be sampled at the end of the switching period and then the first capacitance may be discharged after the sampling. For a preceding or succeeding switching period, a second capacitance may be similarly charged during the on-time and discharged during the off-time. The second capacitance may be sampled at the end of the preceding or succeeding switching period. The control signal ( $V_{control}$ ) may be generated as a linear combination of the sampled voltages of the two capacitances.

The various embodiments described above can be combined to provide further embodiments. These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

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The invention claimed is:

1. A controller for a converter, comprising:

an input configured to receive a first signal representative of a current of a resonant tank of the converter; and an output configured to provide a switching signal for operating the converter, wherein the controller is configured to:

determine a difference between an on-time and an off-time of a command signal representative of the switching signal;

generate a control signal based on the difference between the on-time and the off-time;

compensate the first signal using the control signal; and

generate the switching signal based on the compensated first signal.

2. The controller of claim 1, wherein the controller is configured to compensate the first signal using the control signal by adding the control signal to the first signal to produce the compensated first signal.

3. The controller of claim 1, wherein the controller is configured to generate the control signal as a sum of the difference between the on-time and the off-time in a first switching period and one or more sums of respective differences between the on-time and the off-time in preceding switching periods of the first switching period.

4. The controller of claim 3, wherein the controller is configured to weigh the difference in the first switching period or the differences in the preceding switching periods by a weight in the sum.

5. The controller of claim 1, comprising:

timing logic configured to:

receive a clock signal;

during one of the on-time or the off-time increment a counter every clock cycle of the clock signal;

during the other of the on-time or the off-time decrement the counter every clock cycle of the clock signal; and

determine the difference based on a count maintained by the counter.

6. The controller of claim 1, comprising:

a control node;

a transconductance amplifier configured to:

receive the compensated first signal; and

charge the control node based on the compensated first signal.

7. The controller of claim 6, wherein the control signal mitigates an offset at an input of the transconductance amplifier.

8. A method, comprising:

receiving, by a controller of a converter, a first signal representative of a current of a resonant tank of the converter;

outputting, by the controller, a switching signal for operating the converter; and

balancing, by the controller, diode currents of the converter by at least:

determining a difference between an on-time and an off-time of a command signal representative of the switching signal;

generating a control signal based on the difference between the on-time and the off-time;

compensating the first signal using the control signal; and

generating the switching signal based on the compensated first signal.

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9. The method of claim 8, wherein compensating the first signal using the control signal includes adding the control signal to the first signal to produce the compensated first signal.

10. The method of claim 8, comprising:

generating the control signal as a sum of the difference between the on-time and the off-time in a first switching period and one or more sums of respective differences between the on-time and the off-time in preceding switching periods of the first switching period.

11. The method of claim 10, comprising:

weighing the difference in the first switching period or the differences in the preceding switching periods by a weight in the sum.

12. The method of claim 8, comprising:

receiving a clock signal;

during one of the on-time or the off-time, incrementing a counter every clock cycle of the clock signal;

during the other of the on-time or the off-time, decrementing the counter every clock cycle of the clock signal; and

determining the difference based on a count maintained by the counter.

13. The method of claim 8, comprising:

receiving, by a transconductance amplifier, the compensated first signal; and

charging, by the transconductance amplifier, a control node of the controller based on the compensated first signal.

14. The method of claim 13, wherein the control signal mitigates an offset at an input of the transconductance amplifier.

15. A system, comprising:

a converter; and

a controller including:

an input configured to receive a first signal representative of a current of a resonant tank of the converter; and

an output configured to provide a switching signal for operating the converter, wherein the controller is configured to:

determine a difference between an on-time and an off-time of a command signal representative of the switching signal;

generate a control signal based on the difference between the on-time and the off-time;

compensate the first signal using the control signal; and

generate the switching signal based on the compensated first signal.

16. The controller of claim 15, wherein the controller is configured to compensate the first signal using the control signal by adding the control signal to the first signal to produce the compensated first signal.

17. The controller of claim 15, wherein the controller is configured to generate the control signal as a sum of the difference between the on-time and the off-time in a first switching period and one or more sums of respective differences between the on-time and the off-time in preceding switching periods of the first switching period.

18. The controller of claim 17, wherein the controller is configured to weigh the difference in the first switching period or the differences in the preceding switching periods by a weight in the sum.

19. The controller of claim 15, wherein the controller includes:

timing logic configured to:

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receive a clock signal;  
during one of the on-time or the off-time increment a  
counter every clock cycle of the clock signal;  
during the other of the on-time or the off-time decre-  
ment the counter every clock cycle of the clock 5  
signal; and  
determine the difference based on a count maintained  
by the counter.

**20.** The controller of claim **15**, wherein the controller  
includes: 10

a control node;  
a transconductance amplifier configured to:  
receive the compensated first signal; and  
charge the control node based on the compensated first  
signal. 15

\* \* \* \* \*